Fundamentals of Computer Applications — Combined Assignments

Assignment 1 Computer Characteristics & Generations

🎓 First Year Submission

Computer Characteristics & Hardware Across Generations

Course: **Fundamentals of Computer Applications**

STUDENT NAME	ROLL NO.	COURSE / YEAR
Kshitiz	1826046	BPT 1st Year

1 Four Primary Characteristics of a Modern Digital Computer

Core operational attributes that define modern computing systems

⚡ 1. Speed

MIPS / FLOPS

Modern computers operate at ultra-high speeds, capable of executing billions or trillions of calculations per second. Processing speeds are evaluated in fractions of time such as **microseconds (µs)**, **nanoseconds (ns)**, and **picoseconds (ps)**.

◎ 2. Accuracy

High Precision

Computers perform complex calculations with consistently high precision and near-zero error rates. Errors in output are almost exclusively caused by incorrect human input data or faulty programming, governed by the **GIGO principle (Garbage In, Garbage Out)**.

↻ 3. Diligence

Zero Fatigue

Unlike human workers, computers never suffer from physical tiredness, lack of concentration, or boredom. They can execute millions of complex tasks or repetitive instructions continuously for days with the exact same accuracy and speed.

▣ 4. Versatility

Multi-Tasking

Digital computers are multi-purpose devices capable of executing entirely different categories of tasks seamlessly—from solving mathematical matrices to processing medical scans, driving AI

algorithms, and managing global communications.

2 Key Hardware Technology Across 5 Computer Generations

Evolution of hardware architecture from early vacuum tubes to AI chips

GEN	TIME PERIOD	KEY HARDWARE TECHNOLOGY	PRIMARY FEATURES & MEMORY
1st Gen	1940 – 1956	**Vacuum Tubes**	Large, room-sized systems that consumed high power and produced immense heat. Used magnetic drums for primary memory and machine language instructions.
2nd Gen	1956 – 1963	**Transistors**	Replaced vacuum tubes; vastly smaller, faster, cheaper, and energy efficient. Introduced magnetic core memory and assembly programming languages (FORTRAN, COBOL).
3rd Gen	1964 – 1971	**Integrated Circuits (ICs)**	Semiconductor chips containing hundreds of transistors. Introduced keyboards, monitors, operating systems, and time-sharing processing.
4th Gen	1971 – Present	**Microprocessors (VLSI)**	Very Large Scale Integration compressed millions of transistors onto a single microchip (Intel 4004/CPUs). Sparked personal computing, laptops, and internet networks.
5th Gen	Present & Beyond	**ULSI, AI & Quantum Hardware**	Ultra Large Scale Integration, parallel processing, neural processing units (NPUs), quantum processing chips, natural language understanding, and artificial intelligence.

1st Generation 1940 - 1956

Vacuum Tubes

Relied on glass vacuum tubes for circuitry. Required immense cooling, generated high heat, and calculated using binary code.

2nd Generation 1956 - 1963

Transistors

Invented at Bell Labs. Replaced fragile vacuum tubes, dramatically increasing computing speed and structural reliability.

3rd Generation 1964 - 1971

Integrated Circuits

4th Generation 1971 - Present

Microprocessors (VLSI)

Silicon chips holding multiple miniaturized transistors. Enabled monitors, OS scheduling, and commercial computer software.

Entire Central Processing Unit (CPU) constructed on a single microchip. Powering modern PCs, laptops, and smartphones.

5th Generation Present & Beyond

ULSI, AI & Quantum Hardware

Driven by artificial intelligence, machine learning, neural processors, massively parallel hardware architectures, and emerging quantum computing systems.

Assignment 2

Computer Architecture & Block Diagram

Computer System Architecture & Block Diagram

Interactive Educational Module on Functional Units & Signal Flow

Name: Kshitiz Roll No: 1826046 Class: BPT 1st Year

BUS LEGEND: Data Bus (Bidirectional) | Address Bus (Unidirectional) | Control Bus (Control Signals)

SIMULATIONS: Fetch-Execute Cycle | I/O Data Input | Memory Write Cycle

Interactive Functional Block Diagram

Status: Idle (Click any block or run simulation)

CENTRAL PROCESSING UNIT (CPU)

Brain of Computer

Control Unit (CU)

Decodes instructions & generates control signals across system.

Decoded: FETCH

ALU

Performs arithmetic (+,-,*) & logic (AND, OR) operations.

Op: ADD A, B

Registers Unit (Internal Memory)

PC	IR	MAR	MDR / ACC
0x00A4	LOAD R1	0x00A4	0x002B

SYSTEM BUS INTERCONNECT

↔ Data Bus — Bidirectional

→ Address Bus — Unidirectional

Control Bus Timing & Cmds

MEMORY UNIT Primary Storage

- Cache Memory (L1/L2) — High Speed
- RAM (Main Memory) — 0x00A4: LOAD R1
- ROM / BIOS — Non-Volatile

I/O INTERFACE & PERIPHERALS External World

Input Devices
Keyboard, Mouse, Sensors

Output Devices
Monitor, Printer, Speakers

Click on any functional block (CPU, Memory Unit, I/O Interface, or Buses) above to inspect its internal architecture & role in data flow.

Comprehensive Theoretical Breakdown

Detailed academic notes on Computer Architecture for academic study.

1 System Bus Architecture

A **bus** is a collection of parallel wires transmitting binary electrical signals across system modules.

Data Bus (Bidirectional)

Carries actual data and raw instructions between CPU, Memory, and I/O. Bus width (32-bit, 64-bit) determines word size processing capabilities.

Address Bus (Unidirectional)

Carries physical memory addresses emitted by the CPU to locate specific RAM cells or I/O device ports. Width (e.g., 32 lines = 4GB max memory).

Control Bus (Bidirectional/Multi)

Transmits command and timing signals (e.g., Memory Read/Write, I/O Read/Write, Clock pulses, Interrupt requests, Bus Grant signals).

2 Fetch-Decode-Execute Cycle

The fundamental sequence performed repeatedly by the CPU from power-on until shutdown:

`FETCH` CPU places address from Program Counter (PC) onto Address Bus. Memory returns instruction via Data Bus into Instruction Register (IR). PC increments.

`DECODE` Control Unit (CU) interprets the binary opcode in IR, identifying needed operands and determining required internal control signals.

`EXECUTE` ALU performs requested operation or data is moved to/from memory/registers. Results updated in Accumulator/Registers.

3 I/O & Control Signal Flow

How the CPU coordinates asynchronously with external peripherals and system memory.

I/O Mapping Techniques

Memory-Mapped I/O: Assigns I/O devices address spaces within memory map.
Isolated/Port-I/O: Uses dedicated instruction lines and isolated port address spaces.

Data Transfer Modes

- **Programmed I/O:** CPU continually polls device state.
- **Interrupt-Driven I/O:** Device alerts CPU via interrupt line when ready.
- **Direct Memory Access (DMA):** High-speed transfer directly between I/O and RAM bypassing CPU.

Computer Organization & System Architecture Learning Module

Created for Kshitiz | Roll No: 1826046 | BPT 1st Year

Assignment 3

Low-Level vs High-Level Languages

First Year Submission

Low-Level vs. High-Level Languages

Subject: Computer Applications / Programming Fundamentals

STUDENT NAME	ROLL NO.	COURSE / YEAR
Kshitiz	1826046	BPT 1st Year

Comparative Breakdown by 4 Core Parameters

Detailed examination across key performance and architectural metrics

1. Execution Speed

Low-Level Languages (Machine & Assembly)

Extremely Fast execution. Machine code is executed directly by the CPU without translation delays. Assembly code requires minimal translation and runs near native hardware speeds.

High-Level Languages

Slower execution. Source code must be translated into machine instructions by a Compiler or Interpreter, adding CPU overhead and execution latency.

2. Hardware Dependence

Low-Level Languages (Machine & Assembly)

Highly Hardware-Dependent (Machine Specific). Programs are tailored directly to a specific processor architecture (e.g., x86 vs. ARM) and are non-portable across different systems.

High-Level Languages

Hardware-Independent (Portable). Programs can run on virtually any processor architecture without rewriting, provided a suitable compiler or interpreter exists.

3. Translation Requirements

Low-Level Languages (Machine & Assembly)

No translator for Machine Language; Assembler for Assembly. Machine code (0s and 1s) needs zero translation. Assembly uses a lightweight software called an *Assembler*.

High-Level Languages

Requires Compiler or Interpreter. High-level code (e.g., C++, Python) must undergo complex compilation or step-by-step interpretation to convert into executable machine code.

4. Programming Complexity

Low-Level Languages (Machine & Assembly)

Extremely High Complexity. Requires detailed knowledge of hardware memory, CPU registers, and binary/mnemonic instructions. Highly error-prone and hard to debug.

High-Level Languages

Low Complexity (User-Friendly). Uses human-readable English syntax and mathematical notation. Easy to write, read, maintain, and debug.

Summary Comparison Matrix

At-a-glance side-by-side distinguishability table

PARAMETER	MACHINE LANGUAGE (LOW-LEVEL)	ASSEMBLY LANGUAGE (LOW-LEVEL)	HIGH-LEVEL LANGUAGES
Execution Speed	**Fastest (Direct CPU Execution)**	**Very Fast (Minimal overhead)**	**Slower (Compiler/Interpreter overhead)**
Hardware Dependence	**100% Dependent (System specific)**	**Highly Dependent (Architecture specific)**	**Independent (High Portability)**
Translation	None required (Native	Assembler (Translates mnemonics to machine	Compiler or Interpreter

Requirements	0s and 1s)	code)	required
Programming Complexity	**Extremely Complex (Binary code only)**	**High Complexity (Mnemonics & memory registers)**	**Low Complexity (English-like syntax)**
Memory Efficiency	Maximum control over RAM/Registers	High control over registers	Automated memory allocation
Typical Examples	`10110000 01100001`	`MOV AL, 61h; ADD AL, 1`	`Python, C++, Java, JavaScript`

Assignment 4

Smart Campus Automation (Scanners & Card Readers)

Smart Campus Automation

Optical Data Scanning & Electronic Card Readers System

Name: Kshitiz | Roll No: 1826046 | Program: BPT 1st Year

1. Attendance Simulator | 2. Library Checkout Simulator | 3. System Signal Pipeli

Automated Attendance Management System

Simulate automated student verification using contactless RFID, Smart IC cards, Barcodes, and Batch OMR processing.

Scanner Beep: ENABLED

ELECTRONIC
RFID Tap (13.56 MHz)
NFC / Contactless ID

ELECTRONIC
Smart Card IC
ISO 7816 Contact Chip

OPTICAL
Barcode Scanner
1D Code 128 / CCD Laser

OPTICAL
OMR Sheet Batch
Optical Mark Reader

Mode: RFID Contactless Tap

Status: Ready

Tap Student ID Card near reader

Select Student to Scan:

Kshitiz (1826046)
BPT 1st Year (BPT)

Aarav Sharma
Roll: 1826088 (BPT)

Priya Patel
Roll: 1826012 (BPT)

Simulate Random Scan

TOTAL SCANNED	ON TIME	LATE ENTRY	ATTENDANCE RATE
0	0	0	100%

LIVE VERIFICATION LOG

Clear Log

TIME	STUDENT ID	NAME	METHOD	STATUS
No scan events recorded yet. Perform a scan above.				

Assignment 5

Point-and-Draw Devices vs Graphics Digitizers

Point-and-Draw Devices vs. Graphics Digitizers

Computer Input Hardware: Tracking Mechanics & Pressure Dynamics

Name: Kshitiz Roll No: 1826046 Course: BPT 1st Year

1. Interactive Canvas & CAD Simulator 2. Hardware Tracking Mechanics 3. Tec

Interactive Canvas & CAD Simulator

Test stroke dynamics, pressure curves, relative vs absolute positioning, and CAD vector snapping in real time.

Point-and-Draw (Mouse) Graphics Digitizer (Stylus)

DEVICE CONTROLS MOUSE

Mouse Sensitivity (DPI): 1200 DPI

Relative delta vectors ($\Delta X, \Delta Y$) map mouse velocity to pointer displacement.

Fixed Line Width: 4 px

Mice lack pressure sensors; line weight remains constant (Binary 0/1 click).

CAD Drafting Mode

Smooth Bezier Interpolation

Clear Workspace Canvas

POS: X: 000, Y: 000 MODE: RELATIVE (Delta Vector) PRESSURE: 0 / 4096 RESOL: 1200 DPI

Draw on the canvas above. Switch modes to contrast Mouse jitter with Stylus pressure dynamics.

Canvas Dimensions: 1:1 Scaled Canvas

Computer Input Hardware Demonstration & Simulation System

Designed for Kshitiz | Roll No: 1826046 | BPT 1st Year

Assignment 6

Pen Plotters vs UHD Display + Large-Format Printers

ENGINEERING ASSESSMENT REPORT

Civil Engineering Blueprint Technology Evaluation

Drum/Flatbed Pen Plotters vs. UHD Display & Large-Format Inkjet/LED Printers

Name: Kshitiz Roll No: 1826046 Course: BPT 1st Year

DEFINITIVE SELECTION

Superior Setup:

In modern Architectural, Engineering, and Construction (AEC) workflows, the combination of an **Ultra-High-Definition (4K/8K) Monitor** and a **Large-Format Inkjet or LED Technical Printer** is vastly superior to legacy drum or flatbed pen plotters. While pen plotters once set the benchmark for mechanical vector drafting, modern civil engineering requires complex GIS imagery overlays, BIM 3D color-coding, instant soft-proofing, and high-volume batch submittals that physical pens cannot provide.

2400 × 1200 DPI

Micro-pigment drop placement delivers line weights down to 0.02 mm with full color fill support.

25 Sec / ARCH D Sheet

Multi-page digital batch processing vs. 20–40 minutes per mechanical pen plot.

~65% Lower TCO

Reduced soft-copy print waste, lower mechanical failure rates, and flexible media handling.

Legacy Drum/Flatbed Pen Plotters

- **Slow Mechanical Bottleneck:** Physical pen motors must physically trace every line vector sequentially

sequentially.

- **No Raster / Shading Support:** Cannot render aerial drone topographical photos, satellite overlays, or 3D shaded BIM models.
- **High Mechanical Wear:** Friction pens dry out, skip lines, bleed on media, and require carousel motor calibration.

Modern UHD Monitor + Large-Format Printer

- **Instant Digital Review:** 4K/8K displays allow CAD engineers to soft-proof fine details at 1:1 scale before burning paper/ink.
- **High-DPI Micro-Pigment Ink:** Precise rasterization renders micro-fine lines, solid fills, hatch gradients, and GIS imagery.
- **Field & Regulatory Utility:** Water/UV-resistant pigment hardcopies for wet field environments + digital PDF archiving.

Line Resolution & Drafting Accuracy Analysis

Comparing continuous pen vectors vs. micro-pigment drop placement & UHD display pixels

Drum/Flatbed Pen Plotters

Pen plotters physically drag ink-fed pens across vellum or paper using X-Y stepper motors.

- **Continuous Vector Lines:** Smooth arcs without pixelation step-effects.
- **Line Weight Limitation:** Line width is tied to fixed physical nib sizes (e.g., 0.13mm, 0.25mm, 0.50mm). Changing weights requires mechanical pen carousel switches.
- **Ink Bleeding & Smudging:** Media porosity causes liquid ink to bleed, broadening fine lines beyond tolerance.
- **Zero Solid Fills:** Shading or hatching requires pen cross-hatching, taking dozens of minutes.

UHD Monitor + Large-Format Printer

Combination of GPU-accelerated 4K/8K displays and high-precision piezoelectric inkjet printheads.

- **2400 × 1200 DPI Precision:** Drops as small as 3.5 picoliters create crisp line weights as fine as **0.02 mm**.
- **Color-Coded Layering:** Supports unlimited full-color RGB/CMYK line weights for complex BIM structural cross-sections.
- **Sub-Pixel Anti-Aliasing:** 4K/8K displays (140–220 PPI) render vectors with zero visible staircase artifacts at normal viewing distances.
- **GIS & Raster Support:** Integrates high-resolution satellite imagery directly under CAD line overlays.

Drafting Attribute	Drum / Flatbed Pen Plotter	UHD Display + Large-Format Printer
Min Line Thickness	~0.13 mm (Restricted by physical pen nib)	`0.02 mm (Restricted only by RIP engine)`
Line Accuracy	±0.2% (Subject to motor slippage & paper stretch)	`±0.1% (Laser-calibrated linear encoders)`
Shading & Color Fills	Slow line hatching only; no solid fills	Instant full CMYK/RGB fills & gradient maps
GIS Aerial Imagery	Incapable	Seamless hybrid vector + 300 DPI raster printing

Scalability & Enterprise Workflow Integration

How modern engineering firms scale project submittals and collaborative CAD/BIM reviews

`PHASE 1: REVIEW & REDLINE`

On-Screen Soft Proofing

With 4K/8K UHD displays, structural engineers perform 1:1 scale inspections, digital redlining, and clash

Assignment 7

Vision vs Voice Biometrics in Noisy Environments

Sensory Input Devices Under Adverse Conditions

Critical Analysis: Vision vs. Voice Biometric Authentication in High-Noise Public Environments

Student: Kshitiz Roll No: 1826046 Course: BPT 1st Year

1. Adverse Environment Simulator | 2. Security & Vulnerability Matrix | 3. Hardware

Adverse Environment Simulator & Signal Degrader

Manipulate acoustic noise levels and optical distortion factors to simulate real-time degradation of biometric authentication metrics.

Controlled Lab | Busy Subway Station | Outdoors Stadium / Night

ENVIRONMENTAL FACTORS

STRESS TEST LAB

Acoustic Environment

SNR: +10 dB

Ambient Noise Level ($dB SPL$): 55 dB

Acoustic Interference Type: Stationary Fan

Crowd Speech Babble (Cross-talk)

Optical Environment

500 Lux

Illuminance Level (Lux): 500 Lux

- [ ] Face Mask / Goggles Occlusion
- [ ] Harsh Backlighting / Direct Glare

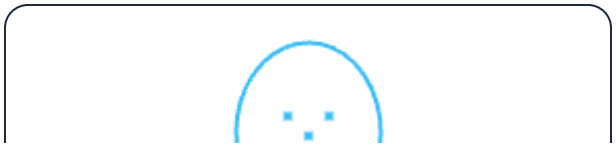
Audio Signal Spectrum (MFCC Input)

CLEAR VOICE



Vision Frame Mesh (Landmark Detection)

68 LANDMARKS





High ambient noise submerges harmonic formant frequencies below the noise floor.



Low illuminance and facial masks eliminate key fiducial landmark coordinates.

BIOMETRIC SYSTEM PERFORMANCE METRICS UNDER STRESS

LIVE EVALUATION

FALSE REJECTION RATE (FRR)		FALSE ACCEPTANCE RATE (FAR)		EQUAL ERROR RATE (EER)		PROCESSING LATENCY	
Vis:	Vox:	Vis:	Vox:	Vis:	Vox:	Vis:	Vox:
0.8%	28.0%	0.00%	1.50%	0.2%	12.1%	150ms	300ms

Nominal performance limits maintained.

Confidence Score: High

Sensory Input Devices Academic Critique & Interactive Simulator

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Assignment 8 Accessible Hands-Free Workstation

Accessible Hands-Free Workstation

NON-VISUAL ARCHITECTURE

Coherent Integration of Non-Visual Inputs & Specialized Synthesizers for Visually Impaired Users

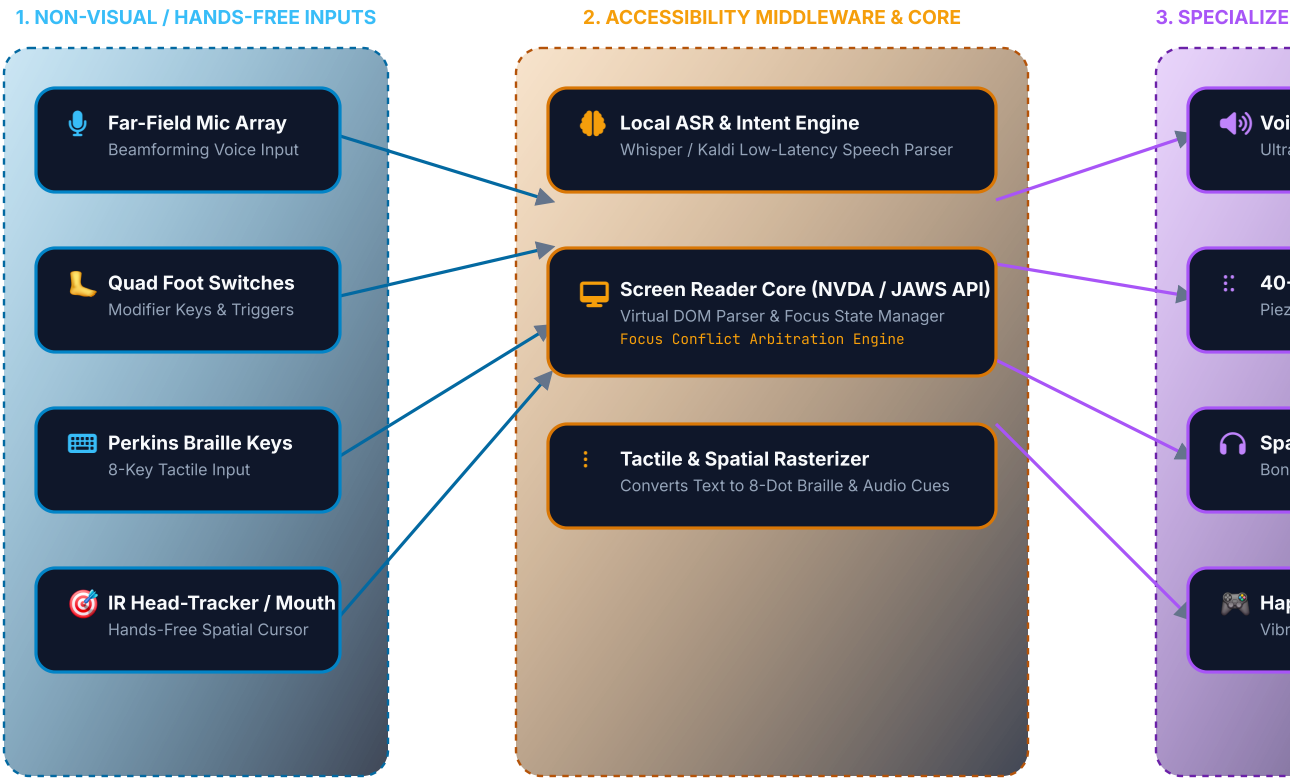
Student: Kshitiz Roll No: 1826046 Course: BPT 1st Year

1. Workstation Topology Visualizer | 2. Audio-Tactile Simulator | 3. Multimodal Data

Workstation Hardware Topology & Architecture

Interactive physical layout showing non-visual inputs, central accessibility middleware, and specialized output synthesizers.

Inputs | Middleware | Outputs



Far-Field Beamforming Microphone Array

INPUT MODALITY

Features a 4-capsule linear array with adaptive DSP noise cancellation. Isolates the blind user's speech voice print from background room noise and TTS synthesized speaker feedback.

Signal Latency	Throughput / Speed	Hands-Free Rating	Protocol Standard
12 ms	160 WPM Voice	10 / 10	USB Audio Class 3

Accessible Hands-Free & Non-Visual Workstation Architecture

Designed for **Kshitiz** | Roll No: **1826046** | **BPT 1st Year**